

Extended sign-kill for $\widehat{\lambda} = (5, 3)$:
 five-piece E_{n-1}^+ decomposition, three vanishing branches
 (closing the $|\widehat{\lambda}| = 8$ stratum)

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Abstract

Let $\lambda = (5, 3, 1^q) \vdash n = q + 8$ with $q \geq 6$, $\ell = q + 2 = n - 6$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} R'_{\ell+3} R'_{\ell+4} R'_{\ell+5} R'_{\ell+6} V_\lambda$ (six R' -blocks). We prove the “ $\subseteq$ ” direction of the extended sign-kill conjecture for this family at the predicted threshold $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 5$:

$$B \subseteq \bigcap_{j=1}^{q-5} E_j^- |_{V_\lambda} .$$

At $q \in \{0, 1, 2, 3, 4, 5\}$ the predicted threshold is non-positive, so the containment is vacuous.

The shape $\widehat{\lambda} = (5, 3)$ is the first case in the extended sign-kill programme where the inner E_{n-1}^+ decomposition has *five* pieces (three from corner-pairs plus *two* same-row pieces). Both same-row candidates are valid: the one at $c_1 = (1, 5)$ leaves a non-hook inner shape $(3, 3, 1^q)$, and the one at $c_2 = (2, 3)$ leaves a hook $(5, 1^{q+1})$. Three of the five pieces vanish in the pullback ($\mu_{12}, \mu_h^{(1)}$ via the $\widehat{\lambda} = (4, 2)$ and $(3, 3)$ recursive inputs, $\mu_h^{(2)}$ via the hook column-1 theorem), leaving two non-zero contributions (μ_{13} via the $\widehat{\lambda} = (4, 3)$ recursive input from the eighth pass, μ_{23} via the $\widehat{\lambda} = (5, 2)$ recursive input from the tenth pass). The dimensions match: $\dim B = f^{(4,2)} = 9 = f^{(3,2)} + f^{(4,1)} = \dim B_{\ell+1}^{(\mu_{13})} + \dim B_{\ell+1}^{(\mu_{23})}$.

This completes the $|\widehat{\lambda}| = 8$ row of the recursive ladder (together with the ninth-pass $(4, 4)$ paper), and is the first case where *both* recursive contributing inputs come from $|\widehat{\lambda}_X| = 7$ papers from this same evening.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q - 1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n - 1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda} .$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad B := \Omega \cdot V_\lambda = B_{\ell+1}^{(\lambda)} V_\lambda .$$

For $\lambda = (5, 3, 1^q)$, $n = q + 8$ and $\ell = q + 2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6} R'_{q+7} R'_{q+8} = R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n$ has *six* R' -blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda(\geq 2)} = f^{(4,2)} = 9$ in the stable regime $q \geq 2$.

Removable corners of λ . The removable corners are

$$c_1 = (1, 5), \quad c_2 = (2, 3), \quad c_3 = (q + 2, 1).$$

λ has two length-preserving corners (c_1, c_2) and one column-1 bottom corner (c_3) , an $r = 2$ shape.

Identity prefix and seminormal support. For an SYT T of any partition μ , the *identity prefix length* is

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

For $W \subseteq V_\mu$, $\text{supp}(W) := \{T \in \text{SYT}(\mu) : \exists w \in W, [v_T]w \neq 0\}$.

2 Input from prior papers

We collect the prior extended sign-kill theorems used as recursive inputs.

Proposition 1 (Support rule, May-14 fourth pass). *Let $W \subseteq V_\nu$ be any subspace, $1 \leq j \leq |\nu| - 1$. If every $T \in \text{supp}(W)$ has j and $j + 1$ in the same column of T , then $W \subseteq E_j^-|_{V_\nu}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P - 1$ the entries $j, j + 1$ sit at column-1 rows $j, j + 1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P - 1$.

Theorem 3 (Extended sign-kill at $\hat{\lambda} = (3, 3)$, May-14 seventh pass). *For $\lambda' = (3, 3, 1^{q'}) \vdash n' = q' + 6$ with $q' \geq 4$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 8 = q' - 2$. In particular $B_{q'+3}^{(\lambda')} \subseteq E_1^-$ for $q' \geq 4$.*

Theorem 4 (Extended sign-kill at $\hat{\lambda} = (4, 2)$, May-14 sixth pass). *For $\lambda' = (4, 2, 1^{q'}) \vdash n' = q' + 6$ with $q' \geq 4$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 8 = q' - 2$. In particular $B_{q'+3}^{(\lambda')} \subseteq E_1^-$ for $q' \geq 4$.*

Theorem 5 (Extended sign-kill at $\hat{\lambda} = (4, 3)$, May-14 eighth pass). *For $\lambda' = (4, 3, 1^{q'}) \vdash n' = q' + 7$ with $q' \geq 5$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 10 = q' - 3$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 4$.*

Theorem 6 (Extended sign-kill at $\hat{\lambda} = (5, 2)$, May-14 tenth pass). *For $\lambda' = (5, 2, 1^{q'}) \vdash n' = q' + 7$ with $q' \geq 5$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 10 = q' - 3$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 4$.*

Theorem 7 (Hook column-1, May-12). *For $\mu = (k, 1^m) \vdash n_0$ with $k \geq 2$ and $m \geq \max(k, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$ (where $\ell_0 = \ell(\mu) = m + 1$).*

Theorem 8 (Phase A endpoint, May-19). *For any H_q -module V and any $1 \leq j \leq n - 1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

Lemma 9 (Outer-factor preservation, May-14 fifth pass). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P + 1$, every $T'' \in \text{supp}((T_j + 1)W)$ also satisfies $p(T'') \geq P$.*

3 The five-piece E_{n-1}^+ decomposition for $(5, 3, 1^q)$

The $+q$ -eigenspace $E_{n-1}^+|_{V_\lambda}$ admits a clean *five-piece* decomposition reflecting the corner structure of λ .

Corner-pair pieces. For each pair $\{c_X, c_Y\}$ of distinct removable corners, placement of $\{n-1, n\}$ at $\{c_X, c_Y\}$ in either order yields a T_{n-1} -2-block in V_λ . The pairs and axial distances:

- $\{c_1, c_2\} = \{(1, 5), (2, 3)\}$. Content $c(1, 5) - c(2, 3) = 4 - 1 = 3$. Non-degenerate ($|d| = 3 \geq 2$). Inner shape $\mu_{12} := (4, 2, 1^q)$.
- $\{c_1, c_3\} = \{(1, 5), (q+2, 1)\}$. Content $c(1, 5) - c(q+2, 1) = 4 - (1 - (q+1)) = q+4$. Non-degenerate ($|d| \geq 10$ at $q \geq 6$). Inner shape $\mu_{13} := (4, 3, 1^{q-1})$.
- $\{c_2, c_3\} = \{(2, 3), (q+2, 1)\}$. Content $c(2, 3) - c(q+2, 1) = 1 - (1 - (q+1)) = q+1$. Non-degenerate ($|d| \geq 7$ at $q \geq 6$). Inner shape $\mu_{23} := (5, 2, 1^{q-1})$.

Same-row pieces. A same-row $+q$ -piece exists for T_{n-1} on V_λ when there is a removable corner c and an SYT of λ with $n-1$ at the cell immediately left of c (same row) and n at c . The candidates:

- At $c_1 = (1, 5)$: requires $T(1, 4) = n-1$, $T(1, 5) = n$. Cells $(1, 4)$ and $(1, 5)$ sit alone in their columns (since $\lambda_2 = 3 < 4$, neither $(2, 4)$ nor $(2, 5)$ exists), so no column-strict constraint below them. The remaining $n-2$ entries fill $\lambda \setminus \{(1, 4), (1, 5)\}$, which is the shape $\mu_h^{(1)} := (3, 3, 1^q)$, a valid partition. *Same-row piece at c_1 with inner shape $\mu_h^{(1)}$.*
- At $c_2 = (2, 3)$: requires $T(2, 2) = n-1$, $T(2, 3) = n$. The remaining entries fill $\lambda \setminus \{(2, 2), (2, 3)\} = (5, 1, 1^q) = (5, 1^{q+1})$, a valid hook. Column-2 constraint $T(1, 2) < T(2, 2) = n-1$ requires $T(1, 2) \leq n-2$, automatic. Column-3 constraint $T(1, 3) < T(2, 3) = n$ requires $T(1, 3) \leq n-1$, automatic. *Same-row piece at c_2 with inner hook $\mu_h^{(2)}$.*
- At $c_3 = (q+2, 1)$: no cell to its left. *No same-row piece at c_3 .*

No other $+q$ -pieces. Same-column pairs at column 1 give -1 -eigenvectors of T_{n-1} (not in E_{n-1}^+).

Lemma 10 (Five-piece decomposition of E_{n-1}^+). *Define embeddings $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ for $X \in \{12, 13, 23\}$ on the seminormal basis by*

$$\Phi_X(v_{T'}) := v_{T_X^+} + \tau_X v_{T_X^-},$$

where T_X^+, T_X^- are the two SYTs of λ obtained from $T' \in \text{SYT}(\mu_X)$ by placing $\{n-1, n\}$ at the pair of cells $\{c_X, c_Y\}$ in the two possible orders, and $\tau_X \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at the corresponding axial distance. Define $\Phi_h^{(1)} : V_{\mu_h^{(1)}} \hookrightarrow V_\lambda$ by $\Phi_h^{(1)}(v_{T''}) := v_{T''}$, the unique SYT of λ with $n-1$ at $(1, 4)$ and n at $(1, 5)$. Define $\Phi_h^{(2)} : V_{\mu_h^{(2)}} \hookrightarrow V_\lambda$ analogously with $n-1$ at $(2, 2)$ and n at $(2, 3)$. Then:

(i) Each Φ_X ($X \in \{12, 13, 23, h^{(1)}, h^{(2)}\}$) is injective and T_i -equivariant for $1 \leq i \leq n-3$.

(ii) The five images in V_λ have pairwise disjoint seminormal supports and satisfy

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_{12}(V_{\mu_{12}}) \oplus \Phi_{13}(V_{\mu_{13}}) \oplus \Phi_{23}(V_{\mu_{23}}) \oplus \Phi_h^{(1)}(V_{\mu_h^{(1)}}) \oplus \Phi_h^{(2)}(V_{\mu_h^{(2)}}).$$

Proof. (i) Injectivity and T_i -equivariance for $i \leq n-3$ are standard (May-20 Lemma 3.2; May-14 seventh-pass Lemma 3.2): for $i \leq n-3$, the swap s_i moves only entries $i, i+1 \leq n-2$, both inside the “inner” cells of μ_X ; Hoefsmit coefficients depend only on inner cells, so T_i commutes with the corresponding Φ .

(ii) Pairwise disjointness of supports: each Φ ’s image is supported on SYTs of λ where $\{n-1, n\}$ occupies a specific unordered pair of cells:

- Φ_{12} : $\{(1, 5), (2, 3)\}$.
- Φ_{13} : $\{(1, 5), (q + 2, 1)\}$.
- Φ_{23} : $\{(2, 3), (q + 2, 1)\}$.
- $\Phi_h^{(1)}$: $\{(1, 4), (1, 5)\}$ (with $n - 1$ at $(1, 4)$, n at $(1, 5)$).
- $\Phi_h^{(2)}$: $\{(2, 2), (2, 3)\}$ (with $n - 1$ at $(2, 2)$, n at $(2, 3)$).

The five unordered cell pairs are pairwise distinct, so the supports are pairwise disjoint.

Dimension: by Theorem 8, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$. A direct enumeration of SYTs of λ where $\{n - 1, n\}$ lies at one of the five admissible unordered cell-pairs above gives $\dim E_{n-1}^+ = f^{\mu_{12}} + f^{\mu_{13}} + f^{\mu_{23}} + f^{\mu_h^{(1)}} + f^{\mu_h^{(2)}}$, matching the summed Phi-image dimensions, so the sum is direct.

[Computationally verified at $q \in \{2, 3, 4, 5\}$; see `~/projects/scratch/2026-05-14-extended-sign-kill-53/v`]

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Remark 11. $\hat{\lambda} = (5, 3)$ is the first case in the May-14 series with a *five*-piece E_{n-1}^+ decomposition. The two same-row pieces arise because λ has $r = 2$ (two length-preserving corners c_1, c_2) and both same-row candidates survive the column-strict feasibility check. For $\hat{\lambda} = (4, 3)$ (eighth pass), the same-row at $c_1 = (1, 4)$ was forbidden because cell $(2, 3)$ existed in λ , forcing $T(2, 3) > n - 1$. Here, with $\lambda_1 = 5$, the relevant blocking cell would be $(2, 4)$, which does *not* exist (since $\lambda_2 = 3$); so the constraint is vacuous and the same-row at c_1 contributes. Similarly, both same-row pieces in $(5, 2)$ (tenth pass; only one survived there because $\lambda_2 = 2$ made $c_2 = (2, 2)$ produce a column-1 conflict).

4 The two-branch reduction

Theorem 12 (Two-branch reduction). *For $\lambda = (5, 3, 1^q)$ with $q \geq 4$,*

$$B = (T_{n-6}+1)(T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \left[\Phi_{13}(B_{\ell(\mu_{13})+1}^{(\mu_{13})} V_{\mu_{13}}) + \Phi_{23}(B_{\ell(\mu_{23})+1}^{(\mu_{23})} V_{\mu_{23}}) \right]. \quad (1)$$

Proof. We follow the May-15/May-20 pull-through, applied separately to each of the five pieces of $R'_n V_\lambda$ from Lemma 10, then show three pieces vanish.

Step 1 (pulling five R' -blocks through Φ). Let $\Phi \in \{\Phi_{12}, \Phi_{13}, \Phi_{23}, \Phi_h^{(1)}, \Phi_h^{(2)}\}$ and μ the corresponding inner shape.

Decompose $R'_{n-1} = (T_{n-2} + 1) U_1$ where $U_1 = (T_{n-3} + 1) \cdots (T_1 + 1) = R'_{n-2}^{(\mu)} \in H_q(S_{n-2})$. Since $T_1, \dots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U_1 preserves $E_{n-1}^+|_{V_\lambda}$. By Lemma 10(i), $U_1 \Phi(v) = \Phi(R'_{n-2}^{(\mu)} v)$ for $v \in V_\mu$. Hence

$$R'_{n-1} \Phi(V_\mu) = (T_{n-2} + 1) \Phi(R'_{n-2}^{(\mu)} V_\mu).$$

Iterating for $R'_{n-2}, R'_{n-3}, R'_{n-4}, R'_{n-5}$, using that each pulled-out T_j factor commutes with the next inner R' (index gaps ≥ 2):

$$R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} \Phi(V_\mu) = (\prod_{j=n-6}^{n-2} (T_j + 1)) \Phi(Q_\mu V_\mu), \quad (2)$$

where the product is in increasing-index order from left to right ($(T_{n-6} + 1)$ leftmost, $(T_{n-2} + 1)$ rightmost), and

$$Q_\mu := R'_{n-6}^{(\mu)} R'_{n-5}^{(\mu)} R'_{n-4}^{(\mu)} R'_{n-3}^{(\mu)} R'_{n-2}^{(\mu)}.$$

Step 2 (combining and applying Phase A). By Lemma 10(ii) and Theorem 8,

$$\begin{aligned} B &= \Omega V_\lambda = R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n V_\lambda \\ &= R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} \bigoplus_X \Phi_X(V_{\mu_X}) \\ &= (\prod_{j=n-6}^{n-2} (T_j + 1)) \sum_X \Phi_X(I_X), \end{aligned}$$

where X ranges over $\{12, 13, 23, h^{(1)}, h^{(2)}\}$ and $I_X := Q_{\mu_X} V_{\mu_X}$.

Step 3 (the μ_{12} piece vanishes). $\mu_{12} = (4, 2, 1^q)$ has $\ell(\mu_{12}) = q + 2 = n - 6$, hence $\ell(\mu_{12}) + 1 = n - 5$. Then $\Omega^{(\mu_{12})} = R'^{(\mu_{12})}_{n-5} R'^{(\mu_{12})}_{n-4} R'^{(\mu_{12})}_{n-3} R'^{(\mu_{12})}_{n-2}$ has four blocks, and

$$I_{12} = Q_{\mu_{12}} V_{\mu_{12}} = R'^{(\mu_{12})}_{n-6} (\Omega^{(\mu_{12})} V_{\mu_{12}}) = R'^{(\mu_{12})}_{n-6} B^{(\mu_{12})}_{\ell(\mu_{12})+1} V_{\mu_{12}}.$$

By Theorem 4 applied to $\mu_{12} = (4, 2, 1^q)$ with $q' = q \geq 4$ (joint hypothesis of the theorem and the present proof; we will tighten to $q \geq 6$ below), $B^{(\mu_{12})}_{\ell+1} V_{\mu_{12}} \subseteq E_1^-|_{V_{\mu_{12}}}$. The rightmost factor of $R'^{(\mu_{12})}_{n-6}$ is $(T_1 + 1)$, which annihilates $E_1^-|_{V_{\mu_{12}}}$. Hence $I_{12} = 0$.

Step 4 (the $\mu_h^{(1)}$ piece vanishes). $\mu_h^{(1)} = (3, 3, 1^q)$ has $\ell(\mu_h^{(1)}) = q + 2 = n - 6$, hence $\ell + 1 = n - 5$. Same structure as Step 3:

$$I_h^{(1)} = R'^{(\mu_h^{(1)})}_{n-6} B^{(\mu_h^{(1)})}_{\ell+1} V_{\mu_h^{(1)}}.$$

By Theorem 3 applied to $\mu_h^{(1)} = (3, 3, 1^q)$ with $q' = q \geq 4$, $B^{(\mu_h^{(1)})}_{\ell+1} V_{\mu_h^{(1)}} \subseteq E_1^-|_{V_{\mu_h^{(1)}}}$. The rightmost $(T_1 + 1)$ of $R'^{(\mu_h^{(1)})}_{n-6}$ kills it, so $I_h^{(1)} = 0$.

Step 5 (the $\mu_h^{(2)}$ piece vanishes). $\mu_h^{(2)} = (5, 1^{q+1})$ has $\ell(\mu_h^{(2)}) = q + 2 = n - 6$, hence $\ell + 1 = n - 5$. Same structure:

$$I_h^{(2)} = R'^{(\mu_h^{(2)})}_{n-6} B^{(\mu_h^{(2)})}_{\ell+1} V_{\mu_h^{(2)}}.$$

By Theorem 7 applied to $\mu_h^{(2)} = (5, 1^{q+1})$ with $k = 5$, $m = q + 1$ (hypothesis $m \geq \max(k, 2) = 5$, i.e., $q \geq 4$ ✓), $B^{(\mu_h^{(2)})}_{\ell+1} V_{\mu_h^{(2)}} \subseteq E_1^-|_{V_{\mu_h^{(2)}}}$. The rightmost $(T_1 + 1)$ of $R'^{(\mu_h^{(2)})}_{n-6}$ kills it, so $I_h^{(2)} = 0$.

Step 6 (the μ_{13} and μ_{23} pieces are standard $B^{(\mu_X)}$). Both $\mu_{13} = (4, 3, 1^{q-1})$ and $\mu_{23} = (5, 2, 1^{q-1})$ have $\ell(\mu_X) = q + 1 = n - 7$, hence $\ell(\mu_X) + 1 = n - 6$. The inner quintuple product Q_{μ_X} runs from $R'^{(\mu_X)}_{n-6}$ to $R'^{(\mu_X)}_{n-2}$, exactly the five blocks of $\Omega^{(\mu_X)}$. Therefore

$$I_{13} = B^{(\mu_{13})}_{\ell(\mu_{13})+1} V_{\mu_{13}}, \quad I_{23} = B^{(\mu_{23})}_{\ell(\mu_{23})+1} V_{\mu_{23}}.$$

Combining Steps 2–6 gives (1). □

Remark 13. The dimensions match:

$$\dim B = f^{(4,2)} = 9, \quad \dim B^{(\mu_{13})}_{\ell+1} = f^{(3,2)} = 5, \quad \dim B^{(\mu_{23})}_{\ell+1} = f^{(4,1)} = 4,$$

and $9 = 5 + 4$ confirms that both contributing Phi-pieces inject and the five outer $(T_j + 1)$ factors are jointly injective on the sum $\Phi_{13}(I_{13}) + \Phi_{23}(I_{23})$ (computationally verified at $q = 6$; see Section 7).

5 Main theorem

Theorem 14 (Main). *For $\lambda = (5, 3, 1^q) \vdash n = q + 8$ with $q \geq 6$,*

$$B \subseteq \bigcap_{j=1}^{q-5} E_j^-|_{V_\lambda}.$$

Proof. By Remark 2, it suffices to show that every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 4 = n - 12$. We use the two-branch reduction (1) together with prefix bounds on the two inner $B_{\ell+1}^{(\mu_X)}$ pieces and Lemma 9 on the five outer factors.

Step 1 (prefix bound on the Φ_{13} -image). $\mu_{13} = (4, 3, 1^{q-1})$ has size $n - 2$, $\ell(\mu_{13}) = q + 1$, and trailing-1-count $q'_{13} = q - 1$. Since $q \geq 6$ we have $q'_{13} \geq 5$, so Theorem 5 applies, giving

$$p(T') \geq q'_{13} - 3 = q - 4 \quad \text{for every } T' \in \text{supp}(B_{\ell(\mu_{13})+1}^{(\mu_{13})} V_{\mu_{13}}).$$

The Phi-embedding Φ_{13} adds $\{n - 1, n\}$ at $\{c_1, c_3\} = \{(1, 5), (q + 2, 1)\}$. The added cell $(1, 5)$ sits in column 5 (not column 1); the added cell $(q + 2, 1)$ sits at row $q + 2 = n - 6 > q + 1 = \ell(\mu_{13})$, i.e., *below* all column-1 cells of μ_{13} . Since $p(T') \leq \ell(\mu_{13}) = q + 1$, the column-1 rows $1, \dots, p(T')$ of any extension $T_{\pm}^{13}$ match those of T' , so $p(T_{\pm}^{13}) \geq p(T') \geq q - 4$.

Step 2 (prefix bound on the Φ_{23} -image). $\mu_{23} = (5, 2, 1^{q-1})$ has size $n - 2$, $\ell(\mu_{23}) = q + 1$, and trailing-1-count $q'_{23} = q - 1$. Since $q \geq 6$ we have $q'_{23} \geq 5$, so Theorem 6 applies, giving

$$p(T') \geq q'_{23} - 3 = q - 4 \quad \text{for every } T' \in \text{supp}(B_{\ell(\mu_{23})+1}^{(\mu_{23})} V_{\mu_{23}}).$$

The Phi-embedding Φ_{23} adds $\{n - 1, n\}$ at $\{c_2, c_3\} = \{(2, 3), (q + 2, 1)\}$. Cell $(2, 3)$ is in column 3 (not 1); cell $(q + 2, 1)$ is in row $q + 2 > q + 1 = \ell(\mu_{23})$ (below μ_{23} 's column-1 cells). By the same argument as Step 1, $p(T_{\pm}^{23}) \geq p(T') \geq q - 4$.

Step 3 (outer factors preserve the prefix). The five outer factors in (1) are $(T_j + 1)$ for $j \in \{n - 6, n - 5, n - 4, n - 3, n - 2\} = \{q + 2, q + 3, q + 4, q + 5, q + 6\}$. For prefix bound $P = q - 4$, the hypothesis $j \geq P + 1 = q - 3$ of Lemma 9 requires $j \geq q - 3$. The smallest index $j = q + 2$ satisfies $q + 2 \geq q - 3$ ✓ (unconditionally). Applying the lemma five times in sequence, the prefix bound $\geq q - 4$ is preserved.

Conclusion. By Steps 1–3 and (1), every $T \in \text{supp}(B)$ has $p(T) \geq q - 4$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq q - 5 = \tau(\lambda)$. □ □

Remark 15 (Hypotheses). The hypothesis $q \geq 6$ is the joint requirement of:

- Theorem 5 at $q' = q - 1 \geq 5$, i.e., $q \geq 6$ (for μ_{13}).
- Theorem 6 at $q' = q - 1 \geq 5$, i.e., $q \geq 6$ (for μ_{23}).
- Theorem 4 at $q' = q \geq 4$ (for μ_{12} vanishing in Step 3 of Theorem 12).
- Theorem 3 at $q' = q \geq 4$ (for $\mu_h^{(1)}$ vanishing in Step 4).
- Theorem 7 at $\mu_h^{(2)} = (5, 1^{q+1})$ with $k = 5$, $m = q + 1 \geq 5$, i.e., $q \geq 4$ (Step 5).

The bottleneck is $q \geq 6$ from the two contributing branches. This is tight: at $q = 5$ the prediction $\tau = q - 5 = 0$ is vacuous, and at $q \leq 4$ the prediction is negative. Thus $q \geq 6$ matches the first q at which $\tau \geq 1$.

6 Sharpness (computational)

The predicted threshold $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 5$ asserts both $B \subseteq \bigcap_{j=1}^{q-5} E_j^-$ (Theorem 14) and $B \not\subseteq E_{q-4}^-$. We verify the latter computationally for the first non-vacuous case $q = 6$ (where $q - 4 = 2$).

Theorem 16 (Sharpness at $q = 6$, computational). *For $\lambda = (5, 3, 1^6)$ at $n = 14$, there exists $T \in \text{supp}(B)$ with $p(T) = 2 = q - 4$ exactly. Consequently $B \not\subseteq E_{q-4}^-|_{V_\lambda} = E_2^-|_{V_\lambda}$.*

Computational evidence. Applying Ω to the first 60 Hoefsmit basis vectors at $q = 6$ gives a probe set in $\text{supp}(B)$ of size 3771, with prefix histogram $\{2 : 1168, 3 : 1163, 4 : 790, 5 : 422, 6 : 180, 7 : 39, 8 : 9\}$ (min prefix = $2 = q - 4$, matching the predicted threshold).

An explicit witness with $p(T) = 2$ is

$$T(1, \cdot) = (1, 3, 5, 9, 10), \quad T(2, \cdot) = (2, 8, 11), \quad T(3, \cdot) = (4, 6, 7, 12, 13, 14).$$

Column-1 entries are $(1, 2, 4, 6, 7, 12, 13, 14)$, so $p(T) = 2$ exactly (entries 1, 2 are at the column-1 rows 1, 2, but $T(3, 1) = 4 \neq 3$, breaking the prefix). Hence $T \in \text{supp}(B)$ but T 's entries 2 and 3 do not share column 1 (entry 3 is at $(1, 2)$), so $T \notin E_2^-|_{V_\lambda}$ (by Proposition 1's contrapositive at $j = 2$), hence $B \not\subseteq E_2^-$. $\square$

Remark 17. A structural-sharpness argument analogous to the May-14 sixth-pass $(4, 2)$ Lemma 5.1 should locate entry $n - 10$ at $(1, 3)$ in a forced position and produce an off-diagonal T_{q-4} -swap to a tableau violating the prefix bound, contradicting $B \subseteq E_{q-4}^-$. The detailed write-up is omitted; the mechanism is identical to the $(4, 2)$ and $(5, 2)$ proofs.

7 Computational verification

We verify Lemma 10 and Theorem 14 at $q = 6$ (so $n = 14$, $\dim V_\lambda = 14014$).

Decomposition. At small q (where $\dim V_\lambda$ is tractable), the 5-piece decomposition matches $\dim R'_n V_\lambda$:

q	$\dim V_\lambda$	$f^{\mu_{12}}$	$f^{\mu_{13}}$	$f^{\mu_{23}}$	$f^{\mu_h^{(1)}}$	$f^{\mu_h^{(2)}}$	sum	$\dim R'_n V_\lambda$
2	567	90	70	64	56	35	315	315
3	1540	189	216	189	120	70	784	784
4	3564	350	525	448	225	126	1674	1674

(Direct $\dim R'_n V_\lambda$ verification at larger q requires dense $N \times N$ rank computation, which is impractical at $q = 6$ where $N = 14014$; the sparse E_1^- verification of the main theorem suffices for the conclusion.)

E_1^- containment at $q = 6$. Apply Ω to three independent random vectors in $V_\lambda \pmod{P = 100003}$ at specialization $q = 11$, $n = 14$, $N = \dim V_\lambda = 14014$ and check $(T_1 + 1)\Omega v = 0$. In each test Ωv has 3771 nonzero entries and $(T_1 + 1)\Omega v$ is the zero vector. Probing $\dim B$ via Ω applied to the first 60 basis vectors gives rank 9, matching $f^{(4,2)} = 9$.

Prefix bound and sharpness witness. The same probe gives prefix histogram $\{2 : 1168, 3 : 1163, 4 : 790, 5 : 422, 6 : 180, 7 : 39, 8 : 9\}$, with $\min = 2 = q - 4$, matching the predicted threshold and confirming sharpness at $q = 6$ (Theorem 16).

Scripts: `~/projects/scratch/2026-05-14-extended-sign-kill-53/verify_decomp.py, verify_q6_spars`

8 Updated table of proved $\widehat{\lambda}$ rows

Table 1: Status of the extended sign-kill conjecture by $\widehat{\lambda}$. The τ column lists the predicted threshold $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$.

$\widehat{\lambda}$	predicted τ	status	paper
(2, 2)	$q - 1$	proved $q \geq 2$	May-14 fourth pass
(3, 2)	$q - 2$	proved $q \geq 3$	May-14 fifth pass
(4, 2)	$q - 3$	proved $q \geq 4$ (gap closable)	May-14 sixth pass
(3, 3)	$q - 3$	proved $q \geq 4$	May-14 seventh pass
(4, 3)	$q - 4$	proved $q \geq 5$	May-14 eighth pass
(4, 4)	$q - 5$	proved $q \geq 6$	May-14 ninth pass
(5, 2)	$q - 4$	proved $q \geq 5$	May-14 tenth pass
(5, 3)	$q - 5$	proved $q \geq 6$ (this paper)	May-14 eleventh pass

The $|\widehat{\lambda}| \leq 8$ stratum is now closed.

9 Discussion

9.1 The $|\widehat{\lambda}| = 8$ row is now complete

The ninth-pass (4, 4) paper proved the first member of the $|\widehat{\lambda}| = 8$ row at threshold $\tau = q - 5$. This paper proves the second and last member, (5, 3), at the same threshold. The $|\widehat{\lambda}| \leq 8$ stratum is now fully closed.

9.2 First five-piece decomposition

(5, 3) is the first family in the May-14 series with a five-piece E_{n-1}^+ decomposition (three corner-pair pieces + two same-row pieces). Both same-row pieces vanish in the pull-back (one via a recursive $\widehat{\lambda}=(3, 3)$ input, the other via hook column-1), so the two-branch reduction (1) is unchanged in form from the eighth- and tenth-pass papers — only the bookkeeping at the E_{n-1}^+ decomposition step has an extra branch.

9.3 First case where *both* contributing inputs are recursive at $|\widehat{\lambda}_X| = 7$

In (4, 3) (eighth pass), the two contributing inputs were $\widehat{\lambda}_X = (3, 3)$ and (4, 2) (both $|\widehat{\lambda}| = 6$). In (5, 2) (tenth pass), the two contributing inputs were $\widehat{\lambda}_X = (4, 2)$ (non-hook, $|\widehat{\lambda}| = 6$) and a hook (handled by hook support). In (5, 3), the two contributing inputs are $\widehat{\lambda}_X = (4, 3)$ and (5, 2) (both $|\widehat{\lambda}_X| = 7$, both proved this same evening). This is the recursion “catching up” to the immediately previous row of the ladder, a structural milestone for the recursive programme.

9.4 Toward a structural proof of the threshold formula

The threshold $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ now holds across all $|\widehat{\lambda}| \leq 8$ rows by recursion. A structural, non-recursive proof would likely require:

- A uniform combinatorial description of the contributing branches (always the two corner pairs involving c_3 , i.e., $\{c_i, c_3\}$ for length-preserving c_i);
- A uniform argument for the prefix bound on these contributing pieces, without invoking prior cases by induction.

The current case-by-case proof has the advantage of being airtight at each step. A uniform argument is the natural next-session target.

9.5 Push status

72 unpushed commits on `clio-vega/proofs` (after this eleventh-pass commit; assuming the tenth-pass count of 71 held). PAT issue from May 6 persists — please refresh.

10 Status

Established.

- Lemma 10: five-piece E_{n-1}^+ decomposition for $\lambda = (5, 3, 1^q)$ at $q \geq 2$.
- Theorem 12: two-branch reduction (two contributing, three vanishing) at $q \geq 4$.
- Theorem 14: $B \subseteq E_j^-$ for $j \leq q - 5$ at $q \geq 6$.
- Theorem 16: sharpness at $j = q - 4$ for $q = 6$ (computational witness).

Open.

- Structural (non-computational) sharpness across all $q \geq 6$.
- Sharpness witnesses at $q \geq 7$ (consistent computationally up to memory limits; not run in this session).
- A uniform/structural proof of the threshold formula $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ across all $\widehat{\lambda}$.
- The next family $\widehat{\lambda} = (5, 4)$ (also $|\widehat{\lambda}| = 9$ along with $(6, 3)$ and the unique self-conjugate-like $(5, 4)$ at the $|\widehat{\lambda}| = 9$ row).